Ashwin Chembu

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PROFILE

Data engineer building Snowflake SQL pipelines, healthcare entity-matching workflows, and traceable quality controls. Connects layered ETL/MDM architecture with API-assisted reconciliation and stakeholder-ready evidence.

TECHNICAL SKILLS

Languages / Platforms: SQL, Python, R, Snowflake, MongoDB, AWS S3, AWS Glue, Pandas, NumPy
Engineering: Layered ETL/MDM, entity resolution, identifier matching, delta logic, source-to-target mapping
Quality / Analysis: Reconciliation, data profiling, QC reports, stewardship review, Power BI, Tableau

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — **South San Francisco, CA | Jun 2024-Present**

- Developed **Snowflake SQL** pipelines across **327K+** healthcare records, combining identifier matching, specialty joins, address comparison, and API connectivity to produce traceable extracts and quality-control reports for review.
- Designed entity-matching and license-reconciliation frameworks across **67K+** records, combining NPI, DEA, HIN, state-license, and fuzzy-search signals to surface candidate matches and exceptions for stewardship review.
- Defined layered **ETL/MDM** architecture from landing and staging through transformation, canonical records, and publication, with data-quality checks, delta logic, and history tracking to preserve traceable changes between sources.
- Built a Node.js provider-lookup workbench with exact identifier search, Jaro-Winkler address ranking, bulk uploads, and session-bound exports; validated authentication, isolation, input checks, and export safety with **10** local tests.
- Translated ambiguous healthcare stakeholder needs into testable rules, API integrations, and source-to-target mappings, documenting acceptance criteria and unresolved decisions so delivery teams could implement scoped changes.

Founder and Lead Developer | Tallyrus — **Jun 2023-Sep 2025**

- Connected Express/Multer upload routes for teacher files, rubrics, and historical essays to AWS S3 storage and **MongoDB** assignment records, making educator materials available to downstream grading and review workflows.
- Integrated LangChain/OpenAI embeddings with NumPy normalization and FAISS similarity search, combining retrieved historical essays with teacher rubrics and prompts in GPT-4o requests to return contextual feedback for educator review.
- Implemented a Flask grading API that validates rubric, essay, prompt, and historical-example inputs before generating JSON feedback, connecting model requests to the review workflow without treating automated output as a final grade.
- Integrated React/TypeScript and Express product workflows with Google OAuth, protected routes, **MongoDB** persistence, and secure sessions, connecting authenticated educator access to document and essay-feedback operations.
- Integrated a pretrained Hugging Face Transformers classifier with PyTorch, adding sentence chunking, CPU/CUDA selection, and length-weighted prediction aggregation behind a Flask API to handle longer text inputs.

Data Analyst / Web Development Lead | Tech4Good at UCSC — **Mar 2023-Jun 2025**

- Analyzed **43,340** anonymized interaction rows across **2,805** sessions with **Python**/Pandas, calculating group active-time measures and building Excel/XLSX exports to turn collaboration records into structured research evidence.
- Built Power BI and Tableau research dashboards alongside **Python,** R, and **SQL** analysis, translating findings into visual summaries for multidisciplinary stakeholders while keeping research interpretation connected to supporting data.
- Separated chat, text-entry, and polling state across **three MongoDB databases**, using change streams, populated message logs, and persisted voter records to synchronize threads, edits, group responses, and reveal/reset operations.
- Implemented **20+** attributed Node/Express REST routes with **MongoDB** persistence and logging for chat, text-entry, and polling workflows, supporting threaded messages, anonymous responses, editing, voting, and admin controls.

PROJECTS

Forge | Personal Fitness Intelligence — **Personal project**

- Integrated Hevy/Oura APIs and Apple Health webhooks with FastAPI/**MongoDB,** combining workout, nutrition, sleep, and recovery records while preferring Oura fields and using Apple Health values to fill missing recovery observations.
- Implemented configurable scheduled Hevy/Oura API pulls, on-demand refresh, MyFitnessPal CSV ingestion, and authorization-reconnection handling, keeping source updates separate from Apple Health push events in the fitness data pipeline.
- Developed an analysis engine that compares actual performance with a **12-week** hypertrophy plan, scoring workout adherence, nutrition compliance, and recovery quality to turn unified fitness records into useful program summaries.

Azure RAG | Document Search Proof of Concept — **Public proof of concept**

- Built a **Python** document-search proof of concept with Azure OpenAI embeddings, LangChain processing, and FAISS similarity search, transforming source text into retrievable vectors that support interactive questions over a document corpus.
- Implemented PDF, text, and Markdown loading with text chunking, vector indexing, and retrieval, connecting relevant source passages to citations so users could inspect the evidence supporting interactive document answers.
- Documented Azure embedding and chat-model deployment configuration for a **30+ document** corpus, providing repeatable setup for ingestion, similarity search, and cited Q&A while keeping the implementation scoped as a proof of concept.

Lume | AI Meeting Intelligence — **Dec 2025-Jan 2026**

- Built a React and Node/Express meeting-intelligence platform with **MongoDB** records and AWS S3 file storage, connecting meeting capture, uploaded documents, transcript-aware search, and project questions in one reusable workflow.
- Built reusable Project Brain knowledge bases from transcripts and seed files, retaining project terminology, relationships, and contextual evidence so later questions could reuse accumulated knowledge across meetings and documents.
- Designed a RAG engine with vector retrieval, intent routing, evidence extraction, and generator/critic refinement, combining relevant meeting and file context into citation-backed answers that users can trace to supporting sources.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7 — **Davis, CA | Graduated Jun 2025**

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu